Learning and Optimizing XGBoost with Planned Experiments

From Blocked Response Surfaces to Explainable Sequential Search

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Abstract

Can planned experiments help us choose useful machine-learning settings, explain their effects, and decide which experiment to run next? We investigate this question through linked studies of XGBoost. Phase A uses California Housing regression to study prediction error and prediction time with a blocked central composite design. It reveals strong curvature, meaningful interactions and sensitivity to random data splits. It also exposes weaknesses: a quadratic response surface can be optimistic at untested configurations, short search-time latency estimates can disagree with verification, and a historically exposed test set cannot provide untouched confirmation. Phase B turns these lessons into a sentiment-classification experiment. Static and sequential central composite designs are compared with multi-objective TPE using shared development folds, equal model-fit budgets and a common warmed prediction-timing operation. Statistical ANOVA effects are compared with Shapley explanations of response models, while separate configurations test surrogate predictions and local word perturbations test explanation fidelity. The sentiment experiment uses GPU training and CPU latency measurement. Mean test macro F1 is 0.87843 for static CCD, 0.87885 for sequential CCD and 0.87843 for MO-TPE. These small observed differences do not establish superiority or equivalence. DOE provides more accurate quadratic predictions at reserved configurations, but the small adaptive extension does not demonstrate a clear benefit over the static design. No method meets the predeclared 162.8 μs latency threshold. The evidence supports DOE as an interpretable and competitive experimental approach, with conclusions limited by the sampled tasks, shared folds, modest replication and historical test exposure.

Keywords: design of experiments, XGBoost, response surfaces, hyperparameter optimization, inference latency, ANOVA, SHAP.

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1 Introduction and research question

A machine-learning model does not have a single automatic setting that is best for every task. In XGBoost, settings called **hyperparameters** control how quickly the model learns, how complex its trees can become, how much data each tree sees, and how strongly learning is regularized. Different settings can change accuracy, prediction time and sensitivity to the data used for evaluation. Our core question is:

> **Can designed experiments find useful XGBoost configurations efficiently, explain why they work, and guide further experiments as reliably as conventional adaptive search?**

The objective is broader than reporting the best accuracy. An experiment should also tell us whether the chosen setting remains useful when the data split changes, whether two settings interact, whether predictions of untested configurations can be trusted, and how much accuracy we exchange for faster predictions. We address these questions in two connected phases. **Phase A** develops the statistical picture on a tabular regression problem. Its successes and failures motivate **Phase B**, which combines statistical response models, model explanations and sequential experimental choices on text classification. The phases form a methodological progression, not a pooled contest between housing and sentiment scores. Their response scales, hardware histories and tasks differ. The original motivation included achieving good selection with fewer evaluations than conventional search. The present evidence primarily compares methods at equal budgets. Equal-budget performance and explanations are useful findings, but they do not by themselves prove substantial evaluation savings.

2 Experimental approach

2.1 Designed experiments and response surfaces

Changing one setting at a time can miss interactions. For example, a small learning rate may work with many trees but look poor with a limited tree budget. Tree depth can improve accuracy while increasing prediction time. A **factorial design** deliberately combines low and high settings so that such joint effects can be studied. A **response** is a measured outcome, such as error or prediction time. **Response surface methodology (RSM)** fits a mathematical approximation linking hyperparameters to those responses. A quadratic surface can represent slopes, interactions and curvature. Curvature means that changing a setting has a different effect in different parts of its range. An interaction means that the effect of one setting depends on another. A **central composite design (CCD)** adds axial points and repeated center points to a factorial core. The axial points help identify curvature. Repeated centers help distinguish systematic structure from repeat-to-repeat variation. A **block** groups evaluations conducted under the same nuisance condition, here a data partition. Modeling block effects helps avoid attributing a favorable split to a favorable hyperparameter. The design is valuable because its observations can

support both selection and explanation. It does not guarantee that a quadratic equation is a faithful description everywhere. That claim must be checked independently.

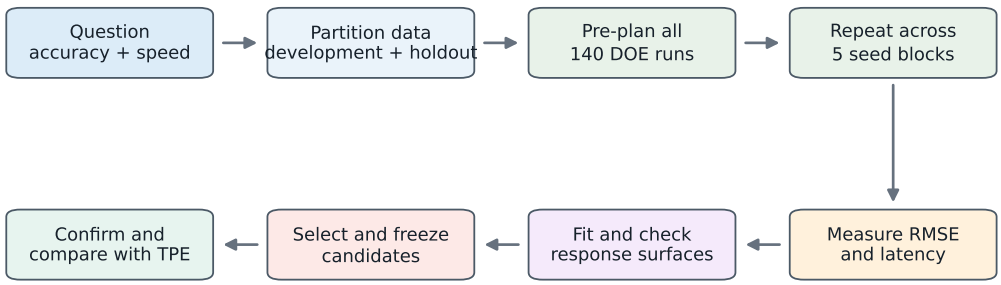


Figure 1: Structure of the housing experiment: a planned design, hierarchical response analysis, compromise selection and independent confirmation.

2.2 Performance measures and multi-objective selection

For housing regression, **root mean squared error (RMSE)** measures prediction error, giving extra weight to large errors. Smaller is better. For sentiment classification, **macro F1** averages the F1 scores of the sentiment classes, giving each class equal weight. Larger is better. The optimization response in Phase B is one minus macro F1, so smaller is better. **Log loss** additionally checks the quality of predicted probabilities and penalizes confident mistakes. **Latency** measures the time needed for a trained model to predict an observation. Lower latency is preferable, but the exact operation matters: constructing an input wrapper, transforming raw text and predicting an already-vectorized document are different tasks. A latency value is meaningful only with its measurement protocol. When no configuration is best on both accuracy and latency, we examine the **Pareto front**: configurations for which no tested alternative improves one objective without worsening the other. **Hypervolume** summarizes the area dominated by that front relative to a declared reference point. Larger hypervolume is preferable under the same objectives and reference. Hypervolumes from different tasks or reference points cannot be compared directly.

3 Phase A: experimental design

3.1 A planned design with stochastic nuisance blocks

The historical design contains 140 fitted evaluations across 5 data-split blocks. It studies learning rate, maximum depth, row subsampling and leaf-weight regularization. A globally randomized, preplanned face-centered CCD supplies the observations. The factorial and augmented analyses are hierarchical analyses of that completed design; they are not evidence that an early result adaptively determined whether the later points were run. XGBoost is trained on development training data and evaluated on development validation data. A separate test partition is used in final evaluation, with an important historical qualification: early versions logged test outcomes during exploratory work. Later access restrictions improve the revised workflow but cannot undo that exposure. The block represents the seeded data partition, not a geographic region. Repeated center evaluations vary the model's stochastic seed while preserving the relevant split. This allows local algorithmic variation to be distinguished from larger changes between configurations and partitions, though those sources should not be interpreted as independent draws from all possible datasets.

Factor	Phase A range	Phase B range
Learning rate	[0.01, 0.30]	[0.05, 0.30]
Tree depth	[3, 9]	[3, 9]
Row subsampling	[0.50, 1.00]	[0.50, 1.00]
Regularization	[0.10, 10.00]	[0.10, 10.00]

Table 1: Hyperparameter ranges in the two phases. Learning rate uses logarithmic coding in Phase A and linear coding in Phase B; regularization uses logarithmic coding in both.

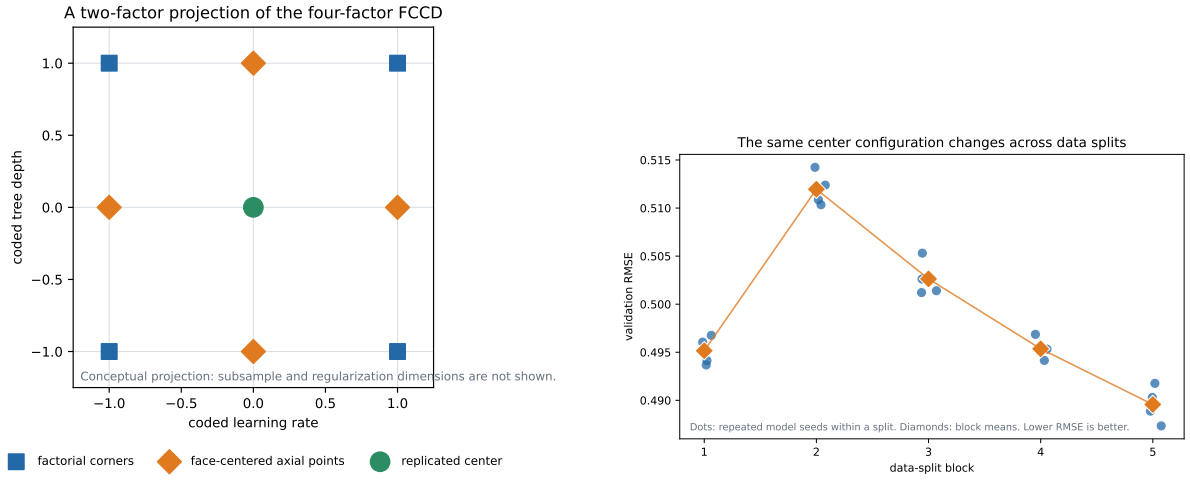


Figure 2: Phase A design and blocking. Left: a projection of the central composite design onto learning rate and depth. Right: center-point outcomes across data-split blocks show why partition effects must be distinguished from hyperparameter effects.

3.2 Screening, response fitting and selection

The first analysis asks whether a flat first-order model is adequate. A strong curvature contrast leads to examining a quadratic response model. The model contains main effects, pairwise interactions and squared terms, together with block effects. **ANOVA** partitions modeled variation and supplies model-conditional tests for these terms. We then use the fitted error and latency surfaces to explore candidate configurations. An accuracy-focused choice minimizes predicted error, while a compromise choice combines the responses through desirability. A fitted stationary point outside the tested region is not treated as an established optimum: it is extrapolation. Candidate choices are checked using separate confirmation observations. The subsequent repeated benchmark compares DOE selections with TPE variants and random search. It is separate from the historical design and confirmation stage. Keeping these campaigns distinct prevents us from mixing validation predictions, confirmation results and final test performance into one apparent experiment.

4 Phase A: results and limitations

4.1 Useful structure, but not a perfect surface

The factorial-to-center curvature statistic is 105432.31. The response clearly departs from a flat plane over the tested range. The block-variance analysis reports an RMSE intraclass correlation of 40.69%. This supports treating data partition as a meaningful nuisance condition rather than ignoring it. However, strong global structure is not the same as local predictive accuracy. The RMSE surface has a lack-of-fit statistic of 62.21. A smooth quadratic equation captures major trends yet misses some systematic response behavior. The curvature contrast uses center-point pure error, whereas the stored lack-of-fit analysis has a different replicate-error decomposition. These denominators must not be conflated, and neither automatically represents uncertainty over all future datasets. One reason the global response is difficult is that 45 of the 140 design runs, or 32.14%, use the lowest learning rate, 0.01. Under the fixed tree budget this region learns poorly. A design that spends many evaluations on that region can produce dramatic curvature without giving equally precise guidance near the best-performing settings. Removing a difficult region after observing it is a diagnostic analysis, not a predeclared new experiment.

4.2 Confirmation revealed boundary optimism

Table 2 compares the response surface with independent confirmation. The compromise candidate's observed validation RMSE falls within its prediction interval. The accuracy-focused candidate is more optimistic: observed error exceeds the upper bound. Its observed-minus-predicted difference is 0.0217 RMSE. The practical lesson is that a mathematically attractive candidate is not yet a verified improvement.

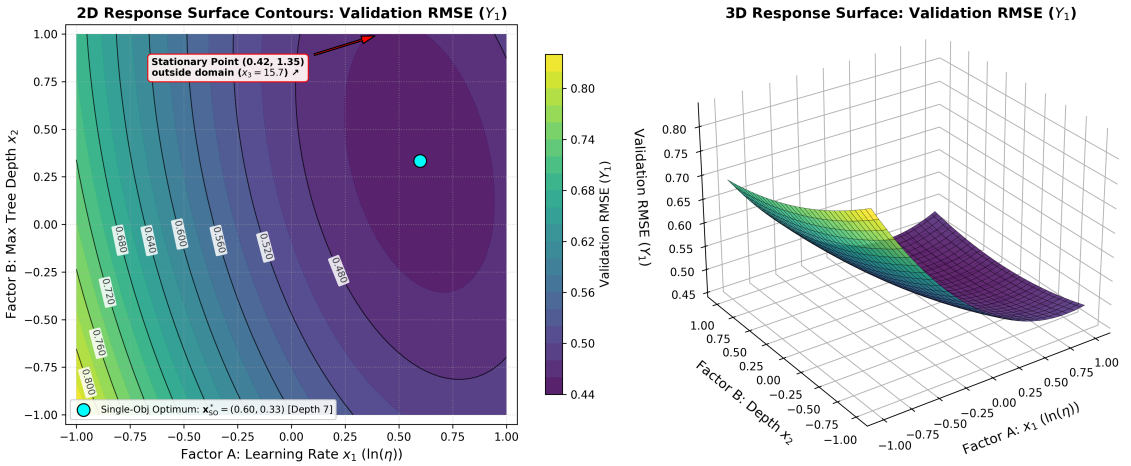


Figure 3: Phase A fitted validation-error surface across coded learning rate and tree depth. Contours and the three-dimensional view show broad curvature, while the out-of-domain stationary point illustrates the danger of extrapolating a fitted optimum.

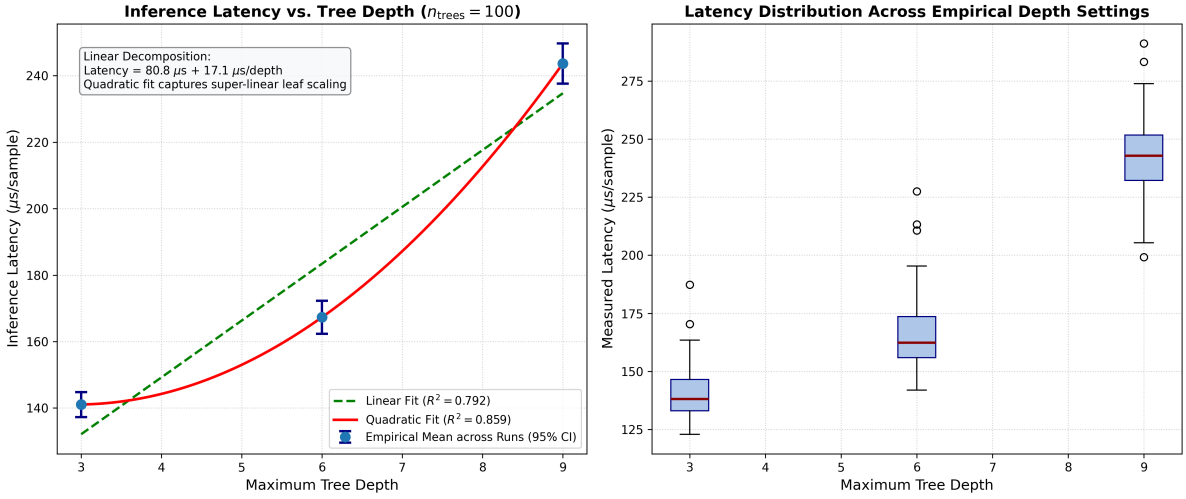


Figure 4: Phase A CPU prediction latency across tree depths. Linear and quadratic fits summarize the change in cost; distributions show variation among measured runs.

Candidate	Predicted	Interval	Observed
Balanced	0.4804	[0.4677, 0.4932]	0.4853
Accuracy	0.4483	[0.4354, 0.4612]	0.4700

Table 2: Phase A validation-RMSE confirmation, reconstructed from saved confirmation runs and their recorded surface predictions. The intervals are model-based prediction intervals, not ranges of every possible future dataset outcome.

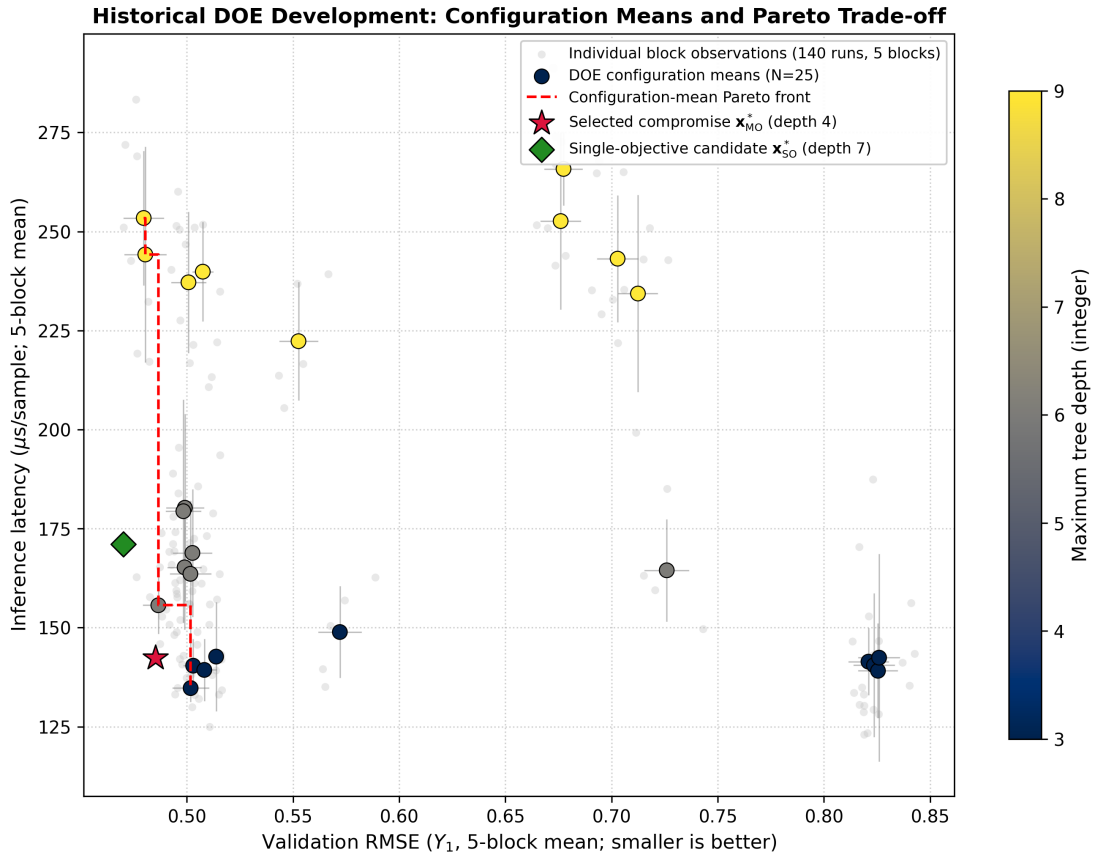


Figure 5: Phase A accuracy–speed trade-off among the initial design configurations. Configuration means and between-block uncertainty distinguish the compromise candidate from the accuracy-focused candidate.

4.3 Optimizer comparison and latency feasibility

Table 3 summarizes the repeated benchmark. Each entry averages selected configurations over search repeats; its error SD describes variability between search-level mean test errors, not the uncertainty of all individual retraining runs pooled together. The accuracy-

Method	Test RMSE	Between-search SD	CPU latency (μs)
Repeated DOE compromise	0.48997	0.00732	122.29
Repeated DOE accuracy	0.46859	0.00000	150.74
MO-TPE	0.49255	0.01080	121.46
Constrained TPE	0.47036	0.00324	144.27
Accuracy-focused TPE	0.46691	0.00116	188.93
Random search	0.46953	0.00207	181.12

Table 3: Phase A repeated-benchmark selected-model results. RMSE is lower-is-better; CPU latency is in microseconds. Differences also reflect the methods' different information allocation and split structures.

focused TPE strategy attains lower observed error at a greater latency cost than the

compromise strategies. This is evidence of a trade-off, not a universal winner. DOE invests observations in structured comparisons and replication; adaptive optimizers spend more on distinct proposed configurations. Latency also exposes a measurement problem. At the predeclared 145 μs threshold, Constrained TPE appears feasible in 20 of 20 searches but only 9 selections remain feasible under benchmark verification. The short online timing and longer final measurement differ in cost and precision. Selection near a noisy boundary can amplify their disagreement. The data do not establish one unique causal explanation for the discrepancy.

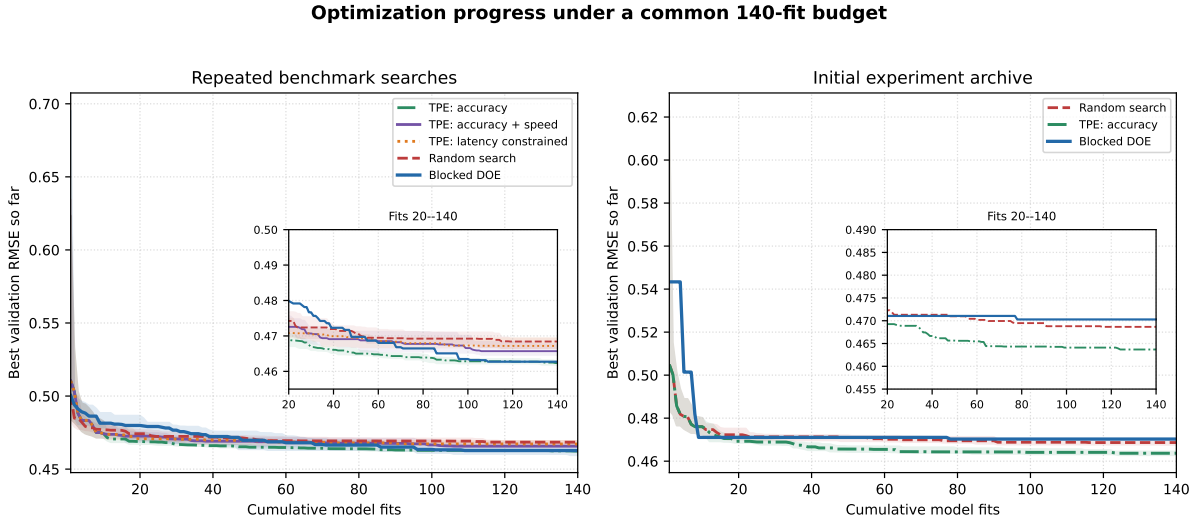


Figure 6: Phase A search trajectories under equal fit budgets. Curves summarize the recorded validation histories; DOE uses multiple validation blocks, so these are descriptive efficiency curves rather than a common-estimand proof of optimizer superiority.

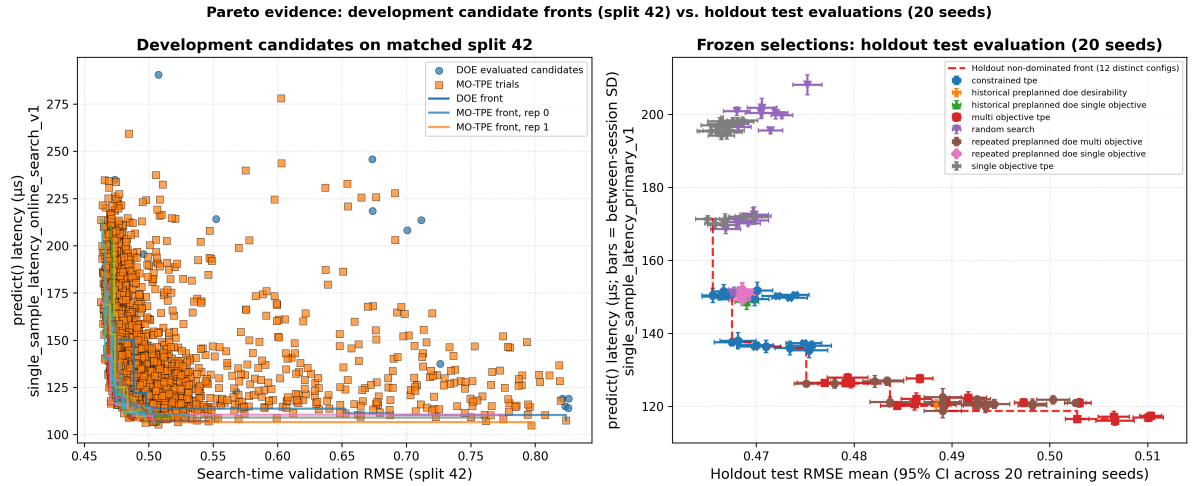


Figure 7: Phase A Pareto comparisons. Development candidate fronts are separated from final evaluated selections, with their uncertainty. A development frontier and a final test frontier describe different evaluation stages.

5 Motivation and research question for Phase B

Phase A establishes that DOE can explain substantial structure and select useful compromise models. It simultaneously identifies reasons to change the next experiment. The learning-rate domain needs development-only calibration; final timing must measure the same operation as search; test history must be stated honestly; and surrogate predictions need checks at genuinely unused hyperparameter coordinates. The next scientific question is therefore:

> **Can an explainable sequential design retain DOE's interpretable effects while improving experimental choices, and can its response predictions and explanations withstand independent checks?**

Moving to sentiment classification gives a different, sparse feature space and allows a second interpretation problem: explaining how selected word-feature changes affect an individual classifier prediction. It does not erase historical test exposure, and it does not turn the two datasets into a representative sample of every machine-learning task. Phase B has several connected objectives: compare ANOVA and SHAP for the same response; predict unseen configurations; test a modest adaptive extension against static CCD; benchmark both against TPE; and evaluate local word-response explanations. These objectives must remain distinguishable. Good classifier accuracy does not automatically establish good explanations, and agreement between explanations does not establish accurate response prediction.

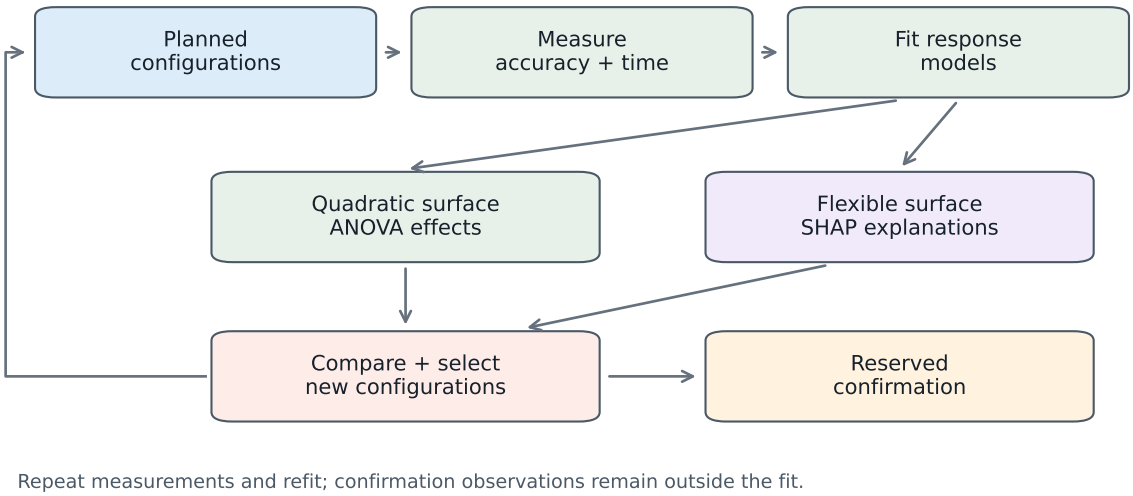


Figure 8: Explainable sequential design: measured responses inform statistical and flexible surrogates; uncertainty and coverage guide new evaluations; reserved coordinates assess predictions. The classifier, response surrogate and local word explainer answer distinct questions.

6 Phase B: experimental methodology

6.1 Data, preprocessing and evaluation

After the recorded text cleaning and deduplication, IMDb supplies 24779 development documents and 24799 test documents. These are the retained counts, rather than the nominal size of the downloaded dataset. TF-IDF converts each document into a sparse vector of word and word-pair weights, using at most 10000 features. Vocabulary and weights are fitted within each training fold, so validation text does not determine that fold's representation. The development pool is partitioned into 5 stratified folds. Every method evaluates a proposed configuration on the same folds and receives the mean response across them. The folds are useful shared nuisance conditions, but their training observations overlap. Different optimizer seeds on those folds do not create independently sampled datasets. All selected configurations are fixed before final test evaluation. Because the test partition has prior exposure, these scores are not fully independent external confirmation. This limits the generalization claim.

6.2 Calibration and computational setting

The lower learning-rate boundary is now 0.05. This avoids the most severe historical setting, but the change alone does not prove adequate learning on text. Development calibration compares tree budgets before fixing the main campaign's capacity. At the low rate, observed calibration macro F1 changes from 0.8049 at 100 trees to 0.8652 at 600 trees. The declared relative-capacity rule selects 600 trees. This is an operational rule within the tested grid, not proof that still larger models cannot improve. Models are trained on an NVIDIA GeForce RTX 5080 accelerator. Prediction latency and local word-response probabilities are measured on CPU. This separates training acceleration from the inference operation being evaluated.

Trees	Learning rate	Validation macro F1	CPU latency (μs)
100	0.175	0.8440	149.1
100	0.050	0.8049	147.6
300	0.175	0.8642	161.2
300	0.050	0.8476	164.4
600	0.175	0.8698	179.1
600	0.050	0.8652	179.1

Table 4: Phase B development calibration. Tree count is selected before the main study; accuracy and timing change with capacity. The timings are calibration observations, not final-model serving guarantees.

6.3 A fair comparison with a small adaptive extension

Static CCD, sequential CCD and MO-TPE each receive 140 actual XGBoost fits per search repeat. The comparison uses 3 repeats. A proposed coordinate is evaluated on every development fold, so model-fit count must not be confused with unique configuration count. Static CCD retains its center replication. Sequential CCD begins with fewer center repetitions and reallocates that budget to adaptive coordinates. A hybrid acquisition rule favors regions with high Gaussian-process uncertainty and separation from existing points. The response models are refitted as observations are added. This is a small sequential extension, not a broad test of every possible adaptive strategy. Its benefit must be judged against the loss of replication as well as any new information. MO-TPE proposes configurations from the same domain and receives the same block-mean loss and latency responses. This arm is multi-objective TPE, not the historical hard-constrained TPE arm. The compromise selection maximizes a dominated rectangle relative to the reference point. It does not require the selected point to satisfy the separately reported latency threshold. The complete Phase B experiment contains 1260 search fits, 540 confirmation fits and 9 final refits, plus the local explanation model: 1810 new model fits in total. Total runtime is 7.40 hours; this excludes the preceding calibration and is not a measured CPU-versus-GPU speedup.

Method	Initial/search fits	Adaptive fits	Total per search
Static CCD	140	0	140
Sequential CCD	130	10	140
MO-TPE	140	0	140

Table 5: Phase B fit allocation per method and search repeat, including all development folds. Sequential CCD reallocates center replication to adaptive points while preserving the total budget. TPE's proposals are reported as search fits.

6.4 The timing protocol and what the threshold means

The common measurement operation uses 100 warmups followed by 500 timed predictions per interface, with CPU affinity and single-thread prediction recorded. The primary operation includes fresh DMatrix construction plus prediction of one already-vectorized document; `inplace_predict` is recorded separately. Raw-text TF-IDF transformation is excluded. The measurement uses the first validation row in each fold. It supports an input-specific microbenchmark, not a latency distribution over representative documents or an end-to-end deployment guarantee. Pinning CPU execution controls one source of variation; it does not eliminate system noise. The latency threshold of 162.8 μs is the median of development calibration timings pooled across tree budgets. It serves as a descriptive feasibility threshold rather than an external deployment requirement. The hypervolume latency reference of 358.3 μs bounds the area used to summarize Pareto quality. Feasibility is assessed against this threshold after measurement. Search and compromise selection optimize the accuracy–latency trade-off without imposing a hard

latency constraint. Because calibration pools different tree capacities, threshold feasibility has a narrower operational interpretation than the observed accuracy–latency trade-off.

7 Phase B: statistical explanation and validation

7.1 ANOVA and SHAP explain the same measured response

For each objective, a blocked quadratic RSM estimates statistical effects. A Gaussian process provides a flexible alternative for predicting configuration-level mean responses. SHAP explains the fitted response model: its inputs are hyperparameters, and its output is predicted classification loss or latency. This is distinct from explaining the sentiment classifier's word features. The analysis uses declared background and explanation coordinates. ANOVA partial sums of squares and Shapley contributions have different meanings and units. ANOVA significance is not SHAP importance; total SHAP importance can include interaction contributions. We compare factor orderings and the pattern of pairwise interactions, supported by curvature profiles and block-specific checks, rather than claiming numerical identity between the methods. Table 6 gives descriptive agreement for classification loss. Factor-rank agreement is more variable than the strong DOE interaction agreement. TPE-based response-model explanations are less consistent in this comparison. That finding is conditional on the sampled points and fitted surrogates; it is not a claim that all TPE explanations are poor.

Method	Factor rank correlation	Interaction correlation
Static CCD	0.800	0.989
Sequential CCD	0.600	0.974
MO-TPE	0.400	0.224

Table 6: Phase B ANOVA versus GP-surrogate SHAP agreement, averaged across searches. The factor column is Spearman rank correlation; the interaction column compares pairwise interaction strengths using correlation across pairwise strengths. These correlations are descriptive and have different estimands.

7.2 New hyperparameter coordinates test response predictions

The experiment reserves 12 coordinates before search, including interior and boundary settings. Their measured responses are excluded from acquisition and surrogate fitting. This tests prediction at new configurations on the existing development folds. Final test documents address a different question: classifier generalization to documents not used for training. The DOE response models predict the reserved loss observations more accurately than the quadratic model fitted to TPE's samples. The static and sequential errors are very close, so this does not demonstrate a clear gain from adaptation. Interval coverage is imperfect, particularly for the TPE-fitted response model. A useful response surface remains an approximation whose uncertainty must be checked. Gaussian-process intervals

Response	Term group	F statistic	Nominal p
Classification loss	Main Effects	69.26	2.2e-30
	Pairwise Interactions	60.85	3.2e-34
	Curvature (Quadratic)	86.00	1.9e-34
	Blocks	18.96	4e-12
CPU latency	Main Effects	434.55	9.4e-71
	Pairwise Interactions	7.21	1.3e-06
	Curvature (Quadratic)	0.22	0.92
	Blocks	2.40	0.054

Table 7: Phase B sequential-CCD ANOVA for a representative search repeat. F statistics describe the fitted model conditional on its sampled coordinates and blocks. Nominal p-values after adaptive sampling are descriptive rather than independent confirmatory tests.

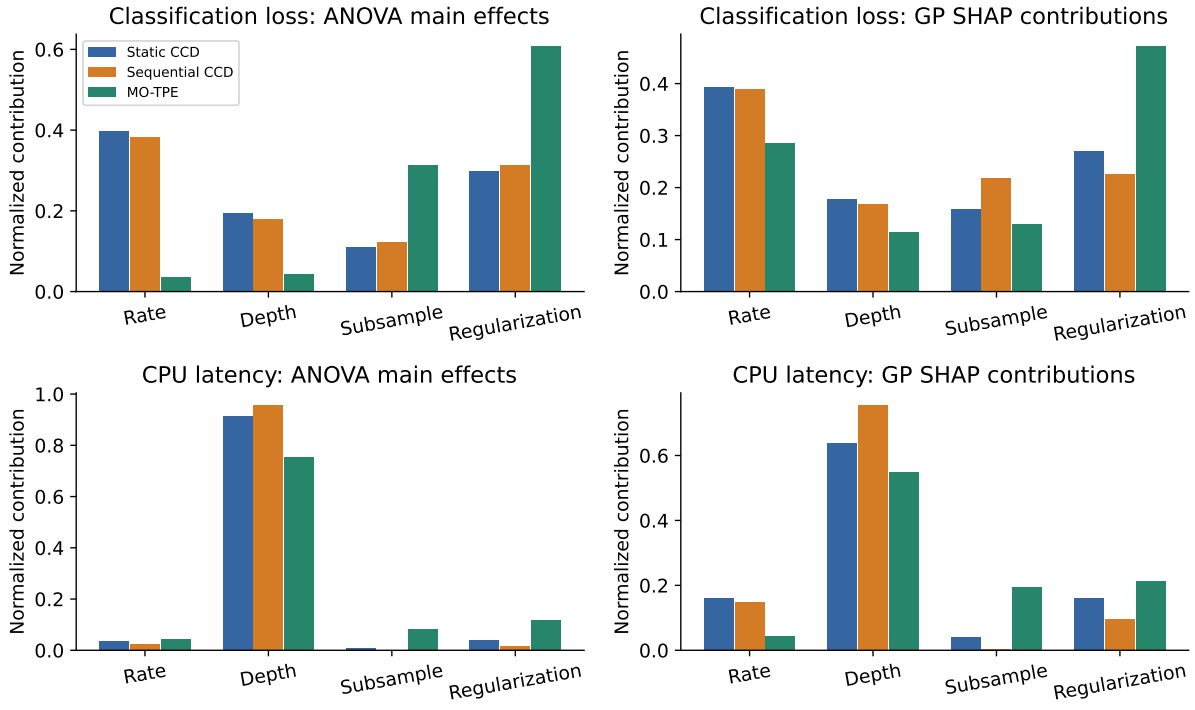


Figure 9: Phase B factor contributions to classification loss and latency. ANOVA main-effect strengths and GP-surrogate SHAP contributions are normalized separately and averaged over repeats. Their relative patterns can be compared; their underlying units and estimands differ.

target block-averaged responses, while the quadratic intervals here are conditional on the modeled blocks. Their coverages cannot be pooled as if they refer to identical future quantities. Neither estimates reliability on all unseen datasets. Formal tests after adaptive sampling and variability across overlapping folds also require cautious interpretation.

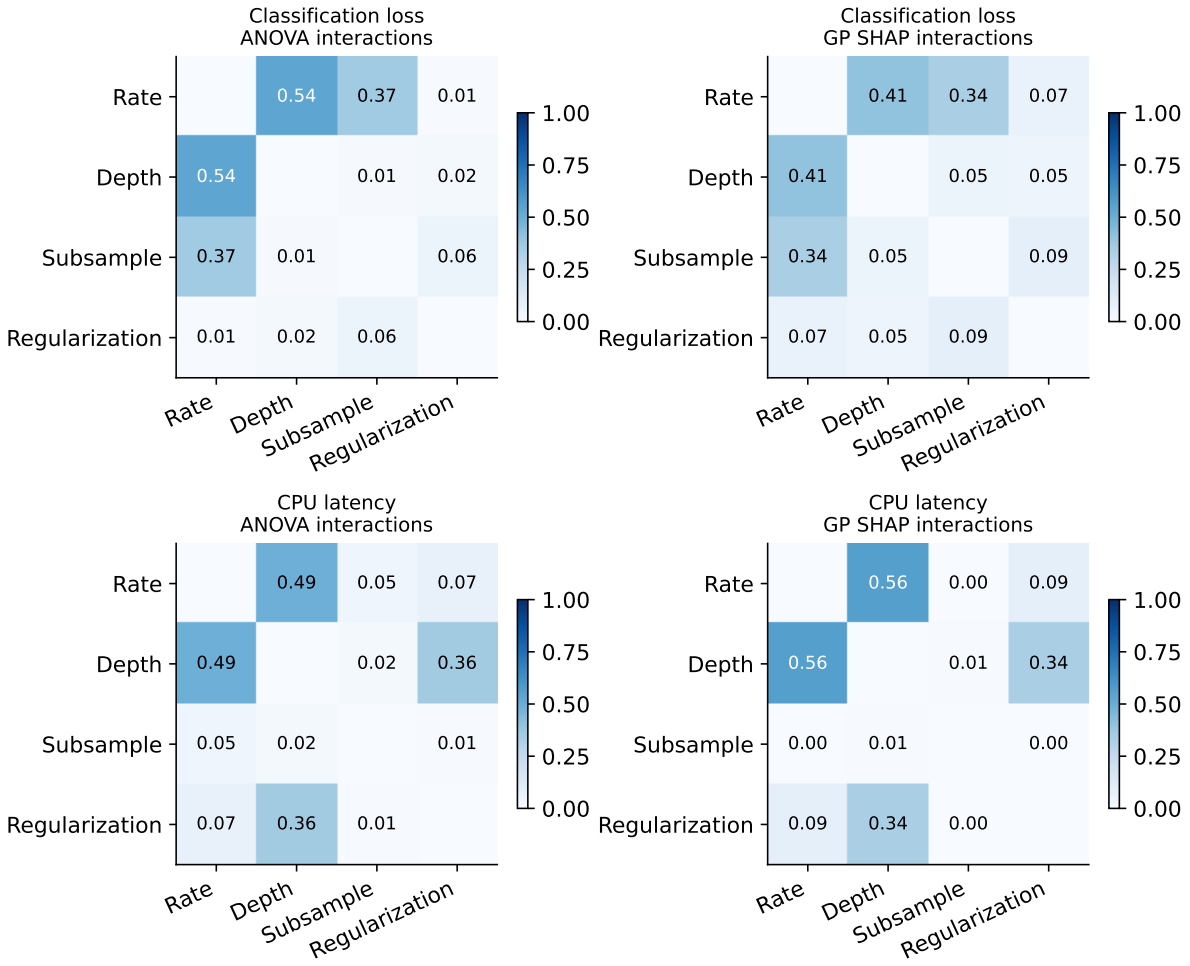


Figure 10: Pairwise interaction patterns for sequential CCD, averaged over repeats. ANOVA and GP SHAP matrices are normalized independently. Agreement concerns which pairs matter, not numerical equality between variance attribution and Shapley contributions.

Method	Response	Pred. RMSE	Mean bias	Coverage	Boundary bias
Static CCD	Loss	0.00343	-0.00046	93.3%	-0.00051
Static CCD	Latency	5.63	-0.75	98.3%	-1.52
Sequential CCD	Loss	0.00335	-0.00014	93.3%	-0.00015
Sequential CCD	Latency	4.78	-0.53	97.8%	-0.66
MO-TPE	Loss	0.00643	0.00133	81.1%	0.00184
MO-TPE	Latency	22.64	-3.25	88.9%	-0.60

Table 8: Phase B quadratic-surrogate confirmation at reserved coordinates. Mean bias is observed minus predicted response. Boundary bias isolates observations near domain limits. Loss errors are in classification-loss units; latency errors are in microseconds.

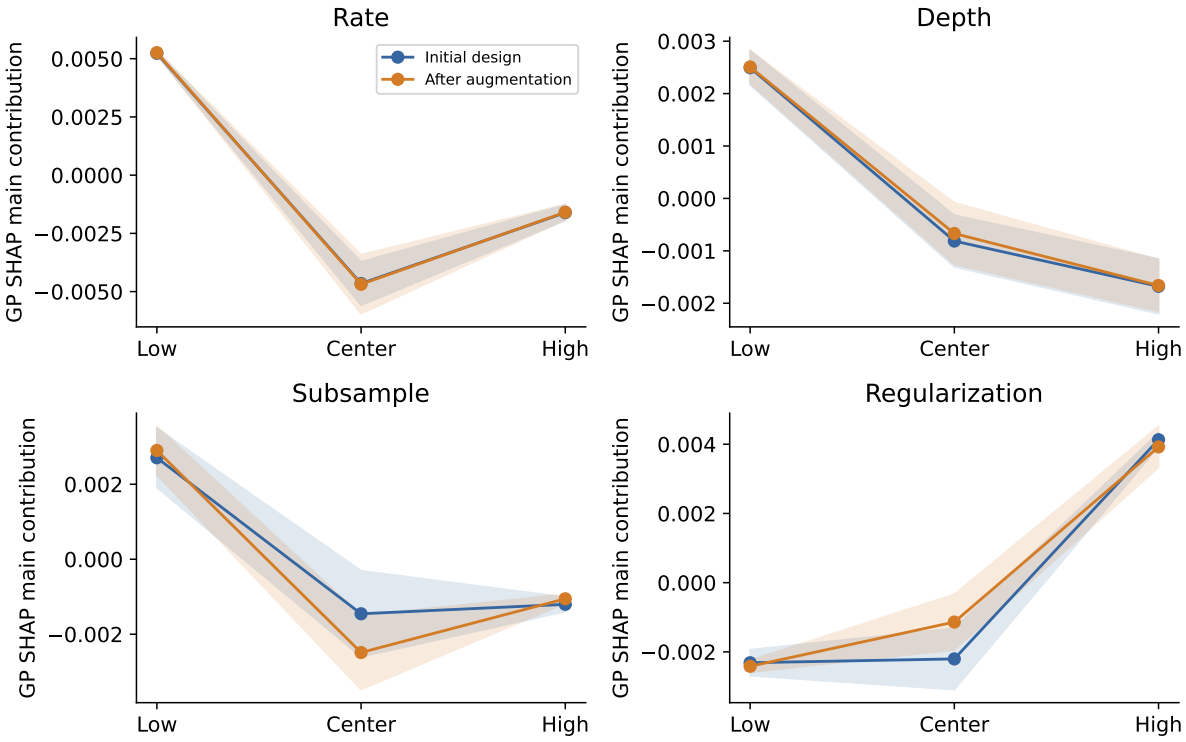


Figure 11: Nonlinear hyperparameter profiles before and after sequential augmentation. Lines show the mean GP SHAP main-effect contribution at low, center and high settings; bands show variation across search repeats. Curvature is visible as a departure from a straight profile.

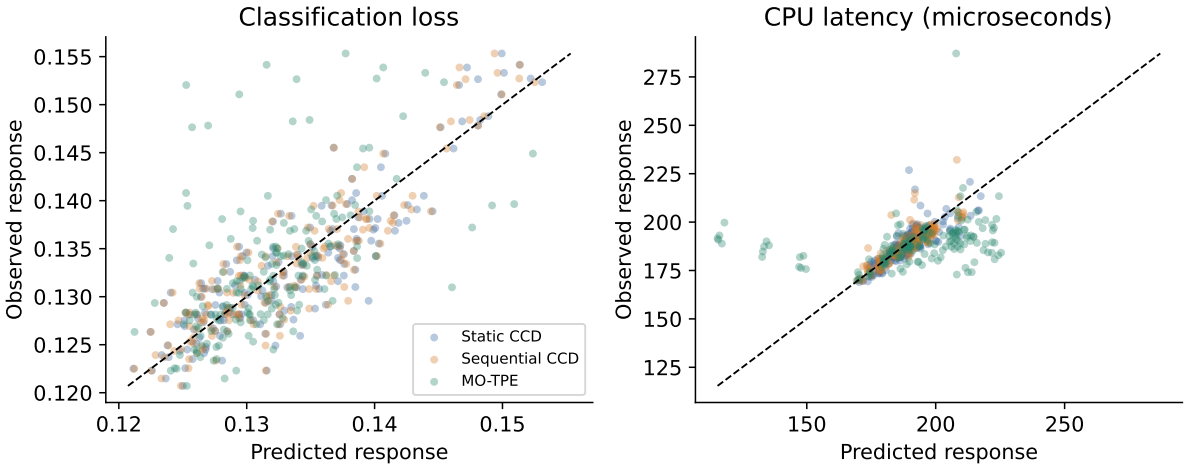


Figure 12: Observed and predicted responses at reserved hyperparameter configurations. The dashed line denotes perfect agreement. Scatter about that line visualizes both prediction error and systematic departures from the response surface.

7.3 Local word explanations answer a separate question

The word study asks how the probability from one fixed sentiment classifier changes when selected word-feature weights are attenuated. A local CCD varies the strongest present unigram features, fits a quadratic probability surface and evaluates fidelity on unused perturbations. Genuine pairwise Shapley values describe the same declared feature-weight game. The study covers 10 eligible development-validation documents chosen by a fixed label-independent rule. The largest document-level confirmation RMSE is 0.0224 probability units. Fidelity varies: some local response functions are simple under these perturbations, while others are less well represented by a quadratic surface. The perturbed inputs are conditional TF-IDF vectors. Other weights, including bigram features, remain fixed and the vector is not renormalized. These are not necessarily grammatical rewritten sentences, and the effects are not causal word effects. LIME is not included as a matched comparison arm, so no comparative superiority claim is made.

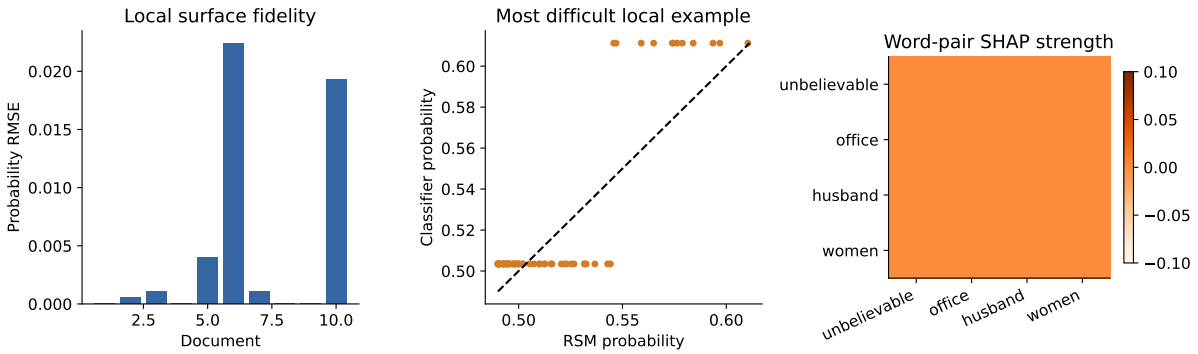


Figure 13: Local word-response results. Left: prediction error for each document on unused feature-weight perturbations. Center: the most difficult local example, document 6, compared with the classifier probabilities. Right: that example's pairwise word-feature Shapley interaction strengths. Word-feature effects are conditional model explanations, not causal language effects.

8 Phase B: comparing model quality, latency and adaptation

Observed test macro F1 is almost the same across methods. The sequential choice has the highest mean in this run, while MO-TPE has the lowest mean selected latency. Static CCD has the highest mean development hypervolume. These small differences do not establish a decisive winner, and absence of a decisive difference does not establish statistical equivalence. No method produces a searched configuration whose block-mean latency meets the recorded 162.8 μs threshold, and none of the final selected models meets it either. This is a negative feasibility result under the chosen definition. The run still supplies valid accuracy and timing observations and useful Pareto comparisons, but it does not demonstrate successful optimization under an externally justified latency requirement. The sequential design does not demonstrate a clear improvement over static CCD on the

Method	Test macro F1	Log loss	CPU latency (μs)	Dev. HV
Static CCD	0.87843 ± 0.00104	0.2879	174.63	162.81 ± 2.76
Sequential CCD	0.87885 ± 0.00072	0.2859	176.12	162.28 ± 2.34
MO-TPE	0.87843 ± 0.00153	0.2864	173.78	161.81 ± 3.10

Table 9: Phase B results. Test metrics average the frozen selected classifier from each search repeat. Development hypervolume summarizes each observed search front under its declared reference. The reported dispersions are sample standard deviations across search repeats. Larger F1 and hypervolume are preferable; smaller log loss and latency are preferable.

present budgets and acquisition rule. It retains explanatory structure and supports a functioning feedback loop, but reallocation of a small amount of replication to infill did not yield an unambiguous performance gain. A broader claim about evaluation savings would require an explicit target-quality analysis with appropriate repetition, not simply equal final fit counts.

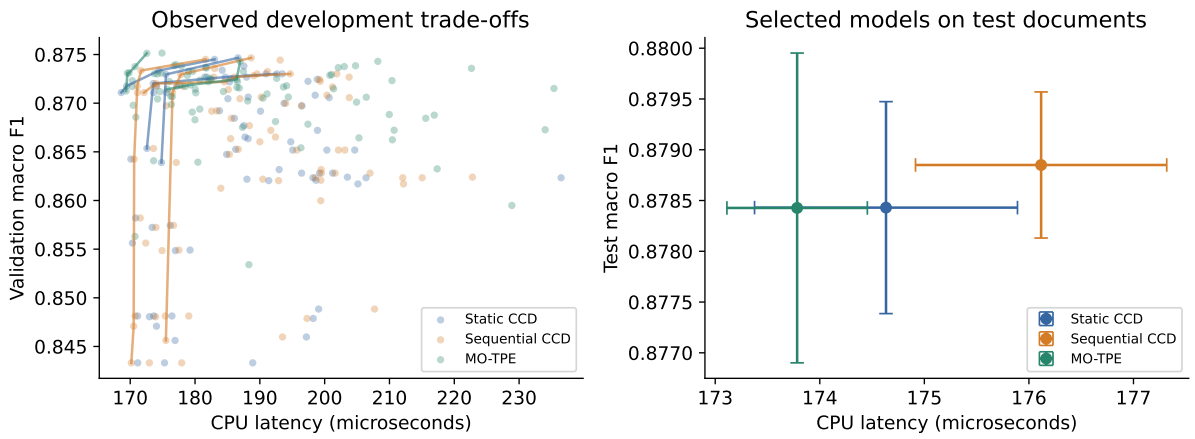


Figure 14: Phase B accuracy–speed outcomes. Left: observed development configurations and their empirical nondominated fronts for each search repeat. Right: frozen selected models on test documents, with sample standard deviations across search repeats. The right panel evaluates selected points, not a complete test-set search front.

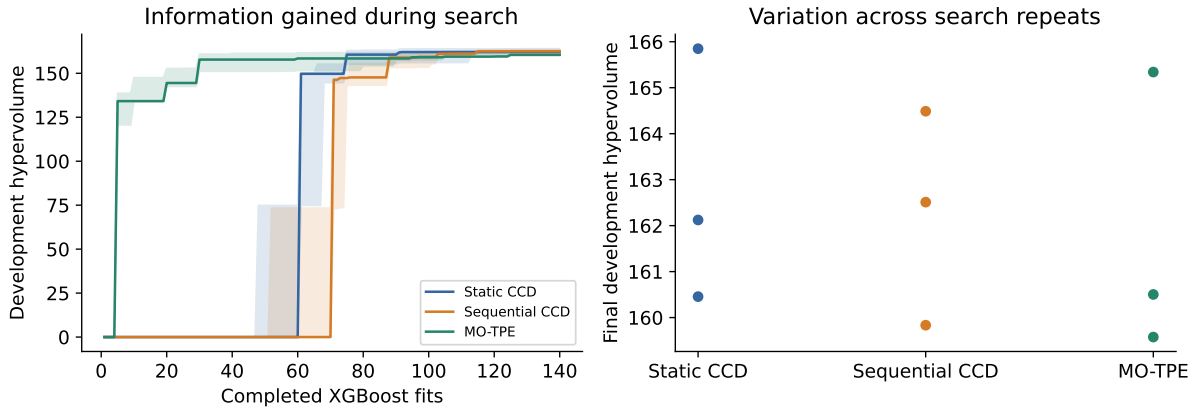


Figure 15: Phase B development hypervolume versus cumulative model-fit count and its final variation across search repeats. The curve uses configurations with complete block evaluations; bands show the interquartile range. Center-repeat updates can change configuration means, so the curve is not a strict best-ever envelope.

9 Discussion

Phase A shows why optimization and explanation need independent validation. Structured data can reveal curvature and interactions while still producing misleading local predictions. A high-quality fitted summary of a broad domain is not automatically accurate near a selected optimum. Timing precision and historical data exposure also affect what comparisons mean. Phase B turns those weaknesses into explicit design choices: development calibration, shared blocks, matched warmed measurements, reserved hyperparameter confirmation, separate final evaluation and reproducible local explanations. Phase B achieves competitive observed model quality under this design. Its strongest explanatory evidence is the agreement of DOE-based interaction descriptions and the relatively accurate quadratic predictions at reserved configurations. Several limitations remain. The classification study uses a limited set of factors, a fixed calibrated tree budget, a small number of search repeats and overlapping folds. The adaptive extension is modest. Its timing panel is narrow, its threshold is internally derived, and the official test set has historical exposure. Housing RMSE and sentiment F1 cannot be pooled into one accuracy result, and hardware-specific latencies cannot establish a general speed law. The two phases therefore provide complementary evidence for a method of investigation, not proof that DOE universally outperforms adaptive optimization or that every explanation reflects a causal mechanism. Unfavorable findings—including failed feasibility and imperfect interval coverage—are part of the scientific outcome.

10 Conclusion

Planned experiments can support useful XGBoost selection while making hyperparameter effects, interactions and uncertainty visible. Phase A established that explanatory structure and exposed the limits of a global response surface and inconsistent verification. Phase B implemented a more explicit explanatory feedback loop and obtained similar

observed classifier quality to TPE, with more accurate DOE quadratic predictions at reserved settings and strong interaction agreement between statistical and surrogate-based explanations. The small sequential extension did not demonstrate a clear advantage over static CCD, and no method met the declared latency threshold. The defensible contribution is a reproducible way to connect design, explanation, acquisition and confirmation while keeping uncertainty and unfavorable evidence visible; stronger efficiency and generalization claims require further independent evidence.

11 Reproducibility and execution

11.1 Obtain the repository and install the environment

The repository is available at https://github.com/ChristosHussein/xgboost-doe-rsm. Use a Python environment compatible with the recorded dependencies and create a separate virtual environment. The following PowerShell commands are intended for the repository root:

```
git clone https://github.com/ChristosHussein/xgboost-doe-rsm.git
cd xgboost-doe-rsm
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
```

The Phase B measurement contract specifies the dependency versions, platform, accelerator and source revision used for the reported experiment. These define the reference environment for reproduction. The repository contains the execution code and published findings. Reconstruction of the reported numerical results additionally requires the corresponding raw experiment directory and its measurement contract. Training again produces a new experimental replication.

11.2 Start with the smoke verification

```
python run_experiment.py --smoke
```

This performs a small execution check of both phases before a full experiment.

11.3 Run Phase A on CPU

The master command runs the revised repeated benchmark rather than silently overwriting the historical housing design:

```
python run_experiment.py --phase a --mode full
```

For explicit output control, the direct runner supports a fresh destination and budget acknowledgement:

```
python phase_a_rsm_blocking/scripts/run_benchmarks.py --mode full --confirm-full-
    budget --output-dir results/housing_new_run
```

The Phase A runner supports `--resume` for interrupted checkpointed runs. Phase A uses CPU training. The historical response-surface study and repeated benchmark are separate experiments.

11.4 Run Phase B with CUDA training or CPU training

For a new CUDA campaign, use the direct Phase B runner so output and calibration paths are explicit:

```
python phase_b_sequential_xai/run_phase_b.py --stage all --dataset full_imdb --data-
    dir phase_b_sequential_xai/data --calibration-dir results/imdb_cuda_new/
    calibration --output-dir results/imdb_cuda_new --findings-dir findings/
    imdb_cuda_new --device cuda
```

For CPU-only training, change the device and use a separate output destination:

```
python phase_b_sequential_xai/run_phase_b.py --stage all --dataset full_imdb --data-
    dir phase_b_sequential_xai/data --calibration-dir results/imdb_cpu_new/calibration
     --output-dir results/imdb_cpu_new --findings-dir findings/imdb_cpu_new --device
    cpu
```

CUDA training requires a CUDA-enabled XGBoost installation and compatible NVIDIA hardware and driver. The measured run records an NVIDIA GeForce RTX 5080; other compatible hardware can run the method but may produce different timings. CUDA accelerates classifier training; surrogate analysis and the prediction-latency operation remain on CPU. The `--training-threads` option controls training-side CPU workers; it does not change the frozen single-thread inference measurement. `--device auto` selects a device automatically. An explicit `cuda` or `cpu` setting makes the computational configuration reproducible.

11.5 Run individual stages and verify saved evidence

The stages `prepare`, `calibrate`, `run` and `generate-findings` can be executed separately. `all` runs them in order, including the word-explanation study. Use a fresh output directory for each replication. Phase B does not provide checkpoint resumption. To verify a completed Phase B study without retraining:

```
python phase_b_sequential_xai/run_phase_b.py --check results/phase_b/imdb
```

To regenerate only its findings, with the raw evidence available:

```
python phase_b_sequential_xai/run_phase_b.py --stage generate-findings --dataset
    full_imdb --output-dir results/phase_b --findings-dir findings/reconstructed_imdb
```

The alternative `yelp` dataset option supports a separate sentiment application. The Phase B results reported here concern IMDb. The measurement contract specifies the

configuration, data partitions, model seeds and computing environment. Recalibration on another machine defines a new replication whose timing results should be reported separately.

A Technical definitions and interpretation

The quadratic response model has the form

$$\widehat{y}(\mathbf{x}, b) = \beta_0 + \sum_i \beta_i x_i + \sum_i \beta_{ii} x_i^2 + \sum_{i<j} \beta_{ij} x_i x_j + \gamma_b.$$

Here x_i are coded hyperparameters and γ_b is a fitted nuisance-block effect. The linear terms describe slopes, squared terms describe curvature, and cross-products describe pairwise interactions. Coding gives the factors comparable numerical scales; it does not make all natural settings equally spaced when a logarithmic transformation is used.

The GP is fitted to coordinate-level mean responses, while the RSM retains block-level observations. Interventional Shapley explanations use the declared background values to replace selected features in coalition evaluations. Main-effect sums of squares, total mean absolute SHAP contributions and pairwise SHAP interaction strengths are distinct summaries. Their agreement is informative only with the model, response and reference distribution stated.

For Phase A benchmark errors, the main table first averages retraining outcomes within each selected configuration, then averages those selected-configuration means across search repeats. It does not treat all retraining rows as independent search runs. Phase B final summaries average the selected classifier from each repeat. Search-front hypervolume is a development measure, while final test scores describe selected points; they do not constitute a completely re-evaluated test front.

An observed fraction inside nominal prediction intervals is an empirical coverage diagnostic. Good coverage can result from wide intervals, and dependent confirmation records do not create independent dataset replications. A nonsignificant difference is not equivalence; a favorable mean is not universal superiority.